Sushobhan Avasthi

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Web Profile
DOB: Jan 10, 1983

Research Interests

Photovoltaics: Lead-halide perovskites; Flexible solar cells; Defect characterization; Si/metal-oxide heterojunctions.
Photodetectors: chalcogenide IR photodetectors; Ge SWIR photodetecotors, Single-photon nanowire photodetectors, X-ray detectors.
Metal-oxides: p-type metal-oxides; Epitaxy of functional oxides; Heterogeneous Integration with Silicon.

Professional Experience

2020–Pres.	**Associate Professor**
	Centre for Nano Science and Engineering (CeNSE), IISc Bangalore
2023–Pres.	**Co-Founder**, ABX3 PV Pvt. Ltd.
	Focus: Commercialization of thin-film perovskite solar technology.
2023–Pres.	**Co-Founder & Director**, Molecular Semiconductors Pvt. Ltd.
	Focus: Production of organic chemicals and precursors for semiconductor research.
2014–2020	**Assistant Professor**
	Centre for Nano Science and Engineering (CeNSE), IISc Bangalore
2011–2014	**Postdoctoral Research Associate**
	Princeton Institute for the Science and Technology of Materials (PRISM), Princeton University, USA
2011–2012	**Co-Founder & CEO**, SuryaTech LLC, USA
	Focus: Commercialization of silicon heterojunction solar technology.

Education

2011	**Ph.D. in Electrical Engineering**, Princeton University, USA
2007	**M.A. in Electrical Engineering**, Princeton University, USA
2005	**B.Tech. in Electrical Engineering**, IIT Kanpur, India

Professional Associations

Senior Member, Institute of Electrical and Electronics Engineers (IEEE)
Young Associate, Indian National Academy of Engineering (INAE)
Member, Materials Research Society of India (MRSI)
Member, Perovskite Society of India (PRSI)

Recognitions

2023	**IISc Award for Excellence in Teaching**
	Recognizes faculty contributions that directly impact student learning and promote a culture of pedagogical excellence at all levels.
2022	**CeNSE Teaching Award**
	Recognizes excellence in teaching and student mentorship within the department for the academic year.
2018	**INAE Young Engineer Award**
	Awarded by the Indian National Academy of Engineering for excellence in research, design, and technology transfer.
2016	**Young Faculty Research Fellowship**
	Awarded under the Visvesvaraya PhD Scheme for Electronics and IT by MeitY, Government of India.
2011	**Prize at 6th Princeton Innovation Forum**
	A business case competition providing seed funding and recognition for solar technology commercialization.
2010	**Edith and Martin B. Stein Solar Energy Innovation Award**
	Awarded by the Global Photonic Energy Corp. for pioneering contributions to solar energy technology.
2005	**Motorola Student of the Year**, IIT Kanpur
	For outstanding academic performance among students in the Electrical and Computer Science departments.

Teaching

NE303 - Semiconductor Process Integration (Credit 2:1)

Offered in the fall semester.

The courses discusses process integration challenges in advanced device microfabrication. I teach the CMOS module where we present the evolution of technology from 1 μm to 2 nm, discussing transitions such as LOCOS to shallow trench isolation, polysilicon to replacement metal gate, and aluminum to Damascene and dual-Damascene interconnects. Other faculty present the evolution of CMOS memory, integrated photonics and microelectromechanical systems. Given the importance of yield, we also have a module on statistical process control. The course has a take-home lab assignments in SEMulator3D®, a process integration emulator, and JMP®, a statistics software.

NE203: Advanced Micro & Nano Fabrication Technology & Processes (Credit 3:0)

Offered in the fall semester.

The course provides an in-depth understanding of the various unit processes in micro & nano fabrication, including crystal growth, doping, vapor deposition, photolithography, wet etching, dry etching, and packaging. The course is accessible to students from diverse backgrounds, such as materials, physics, chemistry, mechanical engineering, and electrical engineering.

NE202: Micro and Nano Fabrication (Credit 0:2)

Offered both semesters.

The course provides hands-on exposure to semiconductor device fabrication. NE202 is a laboratory-only course with one session per week. Students participate in fabrication of solar cells, MOSFETs, and MEMS cantilevers. The course also includes a term project, in which groups of 2-3 students fabricate a device or optimize a unit process.

NE206: Semiconductor Device Physics (Credit 3:0)

Offered in the fall semester.

An graduate level course, NE206 provides an introduction to semiconductor device physics. The focus is on basics like the origin of band-structure, carrier transport, thermal statistics, junctions, defects, and interfaces. Schottky diodes, p-n junction diodes, bipolar junction transistors, and MOS transistors are covered in detail.

This is a fundamental course for anyone interested in electronic devices.

Swayam/NPTEL: Fundamentals of Micro and Nanofabrication
Online course `https://onlinecourses.nptel.ac.in/noc20_bt37/preview`
Intended for students interested in fundamentals of top-down micro and nano device fabrication. The course provides an in-depth understanding of top-down device fabrication. Focus is the unit processes typically used in micro and nano-fabrication of devices. Both concepts and practical aspects are covered. Topics include crystal growth, doping, chemical vapor deposition, physical vapor deposition, photolithography, wet etching, dry etching, and packaging. This is an AICTE approved FDP course.

Peer-Reviewed Publications

[1]T. Saquib, N. Somaiah, S. Kumar, J. Sahoo, R. Muralidharan, S. Avasthi, and D. N. Nath, "Ultrathin n-type doping in β-Ga_2O_3 using SiO_2 solid dopant source", Journal of Vacuum Science & Technology A **44**, 023208 (2026) `10.1116/6.0005185`.

[2]S. Yadav, N. K. Gill, A. A. Mohapatra, A. Dehingia, A. Ghosh, S. Avasthi, N. Chauhan, and S. Patil, "Investigating the limits of PM6:Y6-based organic photodetectors with ternary blend approach", ACS Applied Electronic Materials **8**, 73–80 (2026) `10.1021/acsaelm.5c01513`.

[3]S. Bhardwaj, P. Naik, A. K. Palariya, S. Panda, S. Patil, and S. Avasthi, "Surface and grain boundary passivation using tetraphenylethylene derivative for high-performance perovskite solar cell", Materials Chemistry Frontiers **9**, 2943–2950 (2025) `10.1039/D5QM00255A`.

[4]A. Goswami, S. Dash, S. Avasthi, and V. R. Supradeepa, "Contactless temperature measurement of an in-process silicon wafer using a spectrally shaped supercontinuum source", Opt. Express **33**, 19677–19688 (2025) `10.1364/OE.549560`.

[5]B. Jana, H. K. Gupta, A. Haldar, B. K. Meena, V. Murugan, S. Avasthi, and A. Pandey, "Solution processed tellurium films for photodetection", J. Mater. Chem. C **13**, 17233–17240 (2025) `10.1039/D5TC01755F`.

[6]S. Kundu, Y. Pattipati, K. L. Narasimhan, and S. Avasthi, "How to reliably measure carrier mobility in highly resistive lead halide perovskites with photo-hall experiment", ACS Energy Letters **10**, 3074–3080 (2025) `10.1021/acsenergylett.5c01151`.

[7]B. Sharma, A. A Raj, K. L. Narasimhan, and S. Avasthi, "Enhancing the growth quality of perovskite on MeO-2PACz by introducing a phenethylammonium seed layer", ACS Applied Energy Materials **8**, 17209–17215 (2025) `10.1021/acsaem.5c01999`.

[8]U. Khandelwal et al., "Large electro-opto-mechanical coupling in VO_2 neuristors", Applied Physics Reviews **11**, 021410 (2024) `10.1063/5.0169859`.

[9]M. Mehta et al., "Low interface resistance in epitaxial β-Ga_2O_3 vertical power diodes on silicon (100) using TiN buffer", ACS Applied Electronic Materials **6**, 2084–2092 (2024) `10.1021/acsaelm.4c00258`.

[10]S. K. Parate et al., "Giant electrostriction-like response from defective non-ferroelectric epitaxial $BaTiO_3$ integrated on Si (100)", Nature Communications **15**, 1428 (2024) `10.1038/s41467-024-45903-x`.

[11]A. Pasha et al., "Ionic lockdown: sealing migration channels across device interfaces in mixed halide perovskite solar cells", ACS Energy Letters **9**, 6002–6010 (2024) `10.1021/acsenergylett.4c02546`.

[12]A. A. Raj, S. Bhardwaj, K. L. Narasimhan, and S. Avasthi, "Photoelectromagnetic effect in hybrid lead iodide perovskite", ACS Applied Electronic Materials **6**, 5705–5711 (2024) `10.1021/acsaelm.4c00768`.

[13]S. V. R. Satya, A. R. Singh, S. Vura, S. K. Parate, A. Venugopalarao, S. Raghavan, P. Nukala, and S. Avasthi, "Texture and phase transition hysteresis in epitaxially integrated VO_2 films on TiN/Si(100)", Materialia **34**, 102085 (2024) `10.1016/j.mtla.2024.102085`.

[14]B. Sharma, V. Pawar, S. Patil, and S. Avasthi, "Passivation of low-dimensional quasi-2D Ruddlesden–Popper perovskite solar cells with ionic liquid", ACS Applied Energy Materials **7**, 10163–10170 (2024) `10.1021/acsaem.4c02274`.

[15]B. U, S. Singh, S. Avasthi, P. Vinod, Nagahanumaih, and P. C. Ramamurthy, "Optimization of laser patterning process for eco-friendly tin-halide perovskite solar module – with a nanosecond green laser", Thin Solid Films, 140542 (2024) 10.1016/j.tsf.2024.140542.

[16]A. Vishwakarma, C. S. Gharpure, A. Sugathan, A. Pandey, and S. Avasthi, "Advancements in uncooled bolometer technology: shortwave infrared detection via $CuFeSe_2$ nanocrystal colloidal thin films", physica status solidi (RRL) – Rapid Research Letters **18**, 2300440 (2024) 10.1002/pssr.202300440.

[17]M. Mehta and S. Avasthi, "The possibility of gallium oxide (β-Ga_2O_3) heterojunction bipolar transistors", Physica Scripta **98**, 025013 (2023) 10.1088/1402-4896/acb325.

[18]V. Singh, J. Sinha, and S. Avasthi, "Investigation of vacancies transport in the bilayer of the $Cu_2O_{1\breve{}x}/CuO_{1\breve{}y}$ resistive switching device and effect of growth temperature on memristive switching", ACS Applied Electronic Materials **5**, 6490–6499 (2023) 10.1021/acsaelm.3c00489.

[19]V. Singh, J. Sinha, S. A. Shivashankar, and S. Avasthi, "CVD-deposited Cu_2O thin films with a record Hall hole mobility of 263 $cm^2V^{-1}s^{-1}$ and field-effect mobility of 0.99 $cm^2V^{-1}s^{-1}$", J. Mater. Chem. C **11**, 7356–7366 (2023) 10.1039/D3TC00789H.

[20]J. Singh, P. B. Hegde, S. Avasthi, and P. Sen, "Scalable hybrid antibacterial surfaces: TiO_2 nanoparticles with black silicon", ACS Omega **7**, 7816–7824 (2022) 10.1021/acsomega.1c06706.

[21]K. Kumari, S. Kumar, M. Mehta, A. Chatterjee, S. K. Selvaraja, and S. Avasthi, "Laser-crystallized epitaxial germanium on silicon-based near-infrared photodetector", IEEE Sensors Journal **22**, 11682–11689 (2022) 10.1109/JSEN.2022.3174736.

[22]R. Mukherjee, P. Srivastava, P. Singh, S. Bhardwaj, A. Saha, V. N. Viswanathan, P. C. Ramamurthy, and S. Avasthi, "D-A-D structured polymer interlayer for higher open-circuit voltage and stability in $MAPbI_3$ perovskite solar cells", Journal of Electronic Materials **51**, 6699–6706 (2022) 10.1007/s11664-022-09914-y.

[23]P. Kumar, Seema, M. Gupta, and S. Avasthi, "Fully dense, highly conductive nanocrystalline TiN diffusion barrier on steel via reactive high power impulse magnetron sputtering", Thin Solid Films **722**, 138578 (2021) 10.1016/j.tsf.2021.138578.

[24]K. Kumari, S. Vura, S. Raghavan, and S. Avasthi, "Wafer scale epitaxial germanium on silicon (001) using pulsed laser annealing", Materials Letters **285**, 129208 (2021) 10.1016/j.matlet.2020.129208.

[25]A. Makkaramkott, R. Mukherjee, S. Avasthi, and A. Subramanian, "Ambient prepared mesoporous perovskite solar cells with longer stability", Journal of Electronic Materials **50**, 1535–1543 (2021) 10.1007/s11664-020-08721-7.

[26]A. Nanda, V. Singh, R. K. Jha, J. Sinha, S. Avasthi, and N. Bhat, "Growth-temperature dependent unpassivated oxygen bonds determine the gas sensing abilities of chemical vapor deposition-grown CuO thin films", ACS Applied Materials & Interfaces **13**, 21936–21943 (2021) 10.1021/acsami.1c01085.

[27]D. Paria, V. Vadakkumbatt, P. Ravindra, S. Avasthi, and A. Ghosh, "Unconventional plasmonic sensitization of graphene in mid-infrared", Nanotechnology **32**, 315202 (2021) 10.1088/1361-6528/abf96c.

[28]P. Ravindra, R. Chaudhary, E. Athresh, S. Vura, S. Raghavan, R. Ranjan, K. Majumdar, and S. Avasthi, "Effect of nitrogen annealing on the optoelectronic properties of manganese vanadate", Semiconductor Science and Technology **36**, 055016 (2021) 10.1088/1361-6641/abf0e5.

[29]J. Singh, P. B. Hegde, P. Ravindra, P. Sen, and S. Avasthi, "Ambient light-activated antibacterial material: manganese vanadium oxide ($Mn_2V_2O_7$)", ACS Applied Bio Materials **4**, 6903–6911 (2021) 10.1021/acsabm.1c00605.

[30]V. Singh, J. Sinha, and S. Avasthi, "Two-step process using MOCVD and thermal oxidation to obtain pure-phase Cu_2O thin films transistors", ACS Applied Electronic Materials **4**, 367–374 (2021) 10.1021/acsaelm.1c01050.

[31]V. Singh, J. Sinha, A. Nanda, S. A. Shivashankar, N. Bhat, and S. Avasthi, "Precursor to gas sensor: a detailed study of the suitability of copper complexes as an MOCVD precursor and their application in gas sensing", Inorganic Chemistry **60**, 17141–17150 (2021) 10.1021/acs.inorgchem.1c02410.

[32]P. Kumar, J. M.A., S. Mohan, and S. Avasthi, "Hundred-fold reduction in iron diffusivity in titanium nitride diffusion barrier on steel by microstructure engineering", Thin Solid Films **716**, 138416 (2020) 10.1016/j.tsf.2020.138416.

[33]J. Singh, S. Jadhav, S. Avasthi, and P. Sen, "Designing photocatalytic nanostructured antibacterial surfaces: why is black silica better than black silicon?", ACS Applied Materials & Interfaces **12**, 20202–20213 (2020) 10.1021/acsami.0c02854.

[34]P. Singh, R. Mukherjee, and S. Avasthi, "Acetamidinium-substituted methylammonium lead iodide perovskite solar cells with higher open-circuit voltage and improved intrinsic stability", ACS Applied Materials & Interfaces **12**, 13982–13987 (2020) 10.1021/acsami.0c00663.

[35]P. Kumar and S. Avasthi, "Diffusion barrier with 30-fold improved performance using AlCrTaTiZrN high-entropy alloy", Journal of Alloys and Compounds **814**, 151755 (2020) 10.1016/j.jallcom.2019.151755.

[36]S. Kumar, A. Chatterjee, S. K. Selvaraja, and S. Avasthi, "Two-step liquid phase crystallized germanium-based photodetector for near-infrared applications", IEEE Sensors Journal **20**, 4660–4666 (2020) 10.1109/JSEN.2020.2966759.

[37]A. S. Chouhan, N. Prathibha J., and S. Avasthi, "The dual role of ozone-treated aluminum doped zinc oxide for $CH_3NH_3PbI_3$ solar cells", Organic Electronics **66**, 249–257 (2019) 10.1016/j.orgel.2018.12.030.

[38]R. K. Jha, V. Singh, S. Avasthi, and N. Bhat, "CVD grown cuprous oxide thin film based high performance chemiresistive ammonia gas sensors", IEEE Sensors Journal **19**, 11759–11766 (2019) 10.1109/JSEN.2019.2936223.

[39]P. Ravindra, M. Baral, T. Biswas, M. Nand, S. Jha, E. Athresh, R. Ranjan, M. Jain, T. Ganguli, and S. Avasthi, "Electrical and optical properties of low-bandgap oxide $Zn_2Mo_3O_8$ for optoelectronic applications", Thin Solid Films **677**, 95–102 (2019) 10.1016/j.tsf.2019.03.023.

[40]A. Singh, S. K. Podapangi, and S. Avasthi, "Cation exchange assisted dimensional down-conversion of perovskite thin films using vapor annealing: an interplay of tolerance factor", Integrated Ferroelectric, 10.1080/10584587.2018.1514866 (2019) 10.1080/10584587.2018.1514866.

[41]S. Chaurasia, N. Mohan, S. Raghavan, and S. Avasthi, "Wafer-scale epitaxial germanium (100), (111), (110) films on silicon using liquid phase crystallization", AIP Advances **8**, 075010 (2018) 10.1063/1.5033324.

[42]A. S. Chouhan, E. Athresh, R. Ranjan, S. Raghavan, and S. Avasthi, "$BaBiO_3$: a potential absorber for all-oxide photovoltaics", Materials Letters **210**, 218–222 (2018) 10.1016/j.matlet.2017.09.038.

[43]A. S. Chouhan, N. P. Jasti, and S. Avasthi, "Effect of interface defect density on performance of perovskite solar cell: correlation of simulation and experiment", Materials Letters **221**, 150–153 (2018) 10.1016/j.matlet.2018.03.095.

[44]A. Singh, A. S. Chouhan, and S. Avasthi, "Effect of methylamine vapor exposure and ambient ageing on $Cs_xMA_{1-x}PbI_{3-x}Br_x$ perovskites for improved carrier collection", Materials Research Express **194**, 107–114 (2018) 10.1088/2053-1591/ab1f4f.

[45]V. K. Singh, J. Nagaraju, and S. Avasthi, "Radial junction silicon solar cells with micro-pillar array and planar electrode interface for improved photon management and carrier extraction", Current Applied Physics, 10.1016/j.cap.2018.12.016 (2018) 10.1016/j.cap.2018.12.016.

[46]T. Biswas, P. Ravindra, E. Athresh, R. Ranjan, S. Avasthi, and M. Jain, "Optical properties of $Zn_2Mo_3O_8$: combination of theoretical and experimental study", The Journal of Physical Chemistry C **121**, 24766–24773 (2017) 10.1021/acs.jpcc.7b07473.

[47]A. S. Chouhan, N. P. Jasti, S. Hadke, S. Raghavan, and S. Avasthi, "Large grained and high charge carrier lifetime $CH_3NH_3PbI_3$ thin-films: implications for perovskite solar cells", Current Applied Physics **17**, 1335–1340 (2017) 10.1016/j.cap.2017.07.005.

[48]P. Ravindra, R. Mukherjee, and S. Avasthi, "Hole-selective electron-blocking copper oxide contact for silicon solar cells", IEEE Journal of Photovoltaics **7**, 1278–1283 (2017) 10.1109/JPHOTOV.2017.2720619.

[49]K. A. Nagamatsu, S. Avasthi, G. Sahasrabudhe, G. Man, J. Jhaveri, A. H. Berg, J. Schwartz, A. Kahn, S. Wagner, and J. C. Sturm, "Titanium dioxide/silicon hole-blocking selective contact to enable double-heterojunction crystalline silicon-based solar cell", Applied Physics Letters **106**, 123906 (2015) 10.1063/1.4916540.

[50]K. A. Nagamatsu, S. Avasthi, J. Jhaveri, and J. C. Sturm, "A 12% efficient silicon/PEDOT:PSS heterojunction solar cell fabricated at <100 °C", IEEE Journal of Photovoltaics **4**, 260–264 (2014) 10.1109/JPHOTOV.2013.2287758.

[51]S. Avasthi, W. E. McClain, G. Man, A. Kahn, J. Schwartz, and J. C. Sturm, "Hole-blocking titanium-oxide/silicon heterojunction and its application to photovoltaics", Applied Physics Letters **102**, 203901 (2013) 10.1063/1.4803446.

[52]J. Sturm, S. Avasthi, K. Nagamatsu, J. Jhaveri, W. E. McClain, G. Man, A. Kahn, J. Schwartz, and S. Wagner, "Wide bandgap heterojunctions on crystalline silicon", ECS Transactions **58**, 97–105 (2013) 10.1149/05809.0097ecst.

[53]S. Avasthi, S. Lee, Y.-L. Loo, and J. C. Sturm, "Role of majority and minority carrier barriers silicon/organic hybrid heterojunction solar cells", Advanced Materials **23**, 5762–5766 (2011) 10.1002/adma.201102712.

[54]S. Avasthi, Y. Qi, G. K. Vertelov, Jeffrey Schwartz, A. Kahn, and J. C. Sturm, "Electronic structure and band alignment of 9,10-phenanthrenequinone passivated silicon surfaces", Surface Science **605**, 1308–1312 (2011) 10.1016/j.susc.2011.04.024.

[55]S. Avasthi, Y. Qi, G. K. Vertelov, Jeffrey Schwartz, A. Kahn, and J. C. Sturm, "Silicon surface passivation by an organic overlayer of 9,10-phenanthrenequinone", Applied Physics Letters **96**, 222109 (2010) 10.1063/1.3429585.

[56]S. Shankar, A. M. Tyryshkin, S. Avasthi, and S. A. Lyon, "Spin resonance of 2D electrons in a large-area silicon MOSFET", Physica E: Low-dimensional Systems and Nanostructures **40**, 1659–1661 (2008) 10.1016/j.physe.2007.10.030.

Published Conference Proceedings

[1]A. Vishwakarma, P. Dutta, A. Pandey, and S. Avasthi, "In-situ synthesized Ag_2Te thin films for near-infrared photodetection", in 2025 ieee 10th optoelectronics global conference (OGC) (2025), pp. 134–136, 10.1109/OGC66041.2025.11262134.

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[4]B. Sharma, V. Pawar, and S. Avasthi, "Modification of photophysical properties of NiOx as an htl; transition from 3d to low-dimensional perovskite solar cells", in 2024 ieee 52nd photovoltaic specialist conference (pvsc) (2024), pp. 0447–0447, 10.1109/PVSC57443.2024.10749475.

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[10]S. Kundu and S. Avasthi, "New substituted small a cation(acetamidinium) based tin perovskite solar cell", in 2022 IEEE 49th photovoltaics specialists conference (PVSC) (2022), pp. 1266–1270, 10.1109/PVSC48317.2022.9938906.

[11]S. Kundu, M. Mehta, N. K. L., and S. Avasthi, "Alternating magnetic field hall measurement on low mobility β-ga2o3 thin films", in 2022 IEEE international conference on emerging electronics (ICEE) (2022), pp. 1–3, 10.1109/ICEE56203.2022.10118158.

[12]V. Pawar, A. K. Palariya, N. Anadkat, S. Kumar, and S. Avasthi, "Antisolvent effect on acetamidinium substituted 2d ruddlesden-popper perovskite solar cells", in 2022 IEEE 49th photovoltaics specialists conference (PVSC) (2022), pp. 1133–1135, 10.1109/PVSC48317.2022.9938574.

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Conference Presentations

[1]V. Nagal, V. Adiga, and S. Avasthi, "Revolutionizing photovoltaics: flexible and opaque stainless-steel based perovskite solar cells", 10th International Conference on Composite Materials and Material Engineering (ICCMME 2025), 2025.

[2]V. Nagal and S. Avasthi, "Next-generation photovoltaics: transparent top electrodes and buffer layer in bifacial perovskite solar cells", International Conference on Advances in Sustainable Solutions for Energy Transitions (ASSET 2025), 2025.

[3]R. S. Sandilya, A. Verma, A. R. Singh, S. K. Parate, S. Vura, S. Raghavan, P. Nukala, and S. Avasthi, "Ferroelectric behavior of epitaxial batio3 films grown on tin buffered si using pulsed laser deposition", The 2024 IEEE Ultrasonics, Ferroelectrics, and Frequency Control Joint Symposium, 2024.

[4]A. Verma, S. Bhardwaj, and S. Avasthi, "Quantifying surface passivation of tunnelling oxides by microwave detected photoconductance (mdp)", XXII International Workshop on Physics of Semiconductor Devices (IWPSD-2023), Chennai, India, Dec. 2023.

[5]K. Rudra1, S. Kumar, P. Ravindra, S. Bhat, and S. Avasthi, "Manganese vanadium oxide: first epitaxial thin film and photodetector", 2023 American Crystallographic Association (ACA) Meeting, Baltimore, Maryland, USA, July 2023.

[6]R. S. Sandilya, A. R. Singh, A. Venugopalrao, S. Vura, S. K. Parate, P. Nukala, and S. Avasthi, "Epitaxial vo2 on si 100 via tin buffer layer for cmos applications", MRS Spring Meeting, Materials Research Society, 2023.

[7]A. Vishwakarma, D. Singh, P. Dutta, A. Pandey, and S. Avasthi, "Ag_2Te near-infrared photodetectors", XXII International Workshop on Physics of Semiconductor Devices (IWPSD-2023), 2023.

[8]N. Anadkat and S. Avasthi, "Investigation of defects in acetamidinium-substituted $CH_3NH_3PbI_3$ perovskite solar cells via voltage-dependent thermal admittance spectroscopy", 8th World Conference on Photovoltaic Energy Conversion (WCPEC-8), 2022.

[9]A. A. Raj, S. Bhardwaj, K. Narasimhan, and S. Avasthi, "Photoelectromagnetic effect in perovskite thin films [best poster award]", 8th World Conference on Photovoltaic Energy Conversion (WCPEC-8), 2022.

[10]N. A. S. Kumar, V. Pawar, and S. Avasthi, "Semi-transparent acetamidinium substituted perovskite solar cells", XXI International Workshop on Physics of Semiconductor Devices (IWPSD-2021), 2021.

[11]V. Pawar, N. Anadkat, A. K. Palariya, S. Kumar, and S. Avasthi, "Acetamidinium-substituted 2D Ruddlesden Popper halide perovskite solar cell", XXI International Workshop on Physics of Semiconductor Devices (IWPSD-2021), 2021.

[12]V. Singh, J. Sinha, and S. Avasthi, "Bandgap and defect states analysis using spectroscopy of chemical vapor deposition (cvd) grown Cu_2O", 2021 Fall Meeting of Material Research Society, 2021.

[13]K. Kumari and S. Avasthi, "Pulsed laser crystallization of amorphous germanium thin films over silicon substrate", 10th International Conference on Materials for Advanced Technology (ICMAT), 2019.

[14]S. K. Podapangi, N. P. T., A. Singh, and S. Avasthi, "Spray-coated planar perovskite solar cells processed in ambient on NiOx substrates: an optimization voyage from thin-films to devices", 10th International Conference on Materials for Advanced Technology (ICMAT), 2019.

[15]V. Singh, J. Sinha, and S. Avasthi, "Manipulating oxygen potential to obtain Cu_2O by oxidation and reduction by chemical vapor deposition", 10th International Conference on Materials for Advanced Technology (ICMAT), 2019.

[16]V. Singh, J. Sinha, S. A. Shivashankar, and S. Avasthi, "Effect of oxygen during growth of copper and its oxides thin films using chemical vapor deposition", 2nd International Conference on Nanoscience and Nanotechnology, 2019.

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[18]A. S. Chouhan, N. P. Jasti, and S. Avasthi, "Surface modification of aluminum doped zinc oxide by ozone-gas treatment for perovskite solar cells", International Conference on Perovskite Solar Cells, Photonics and Optoelectronics 2019, 2018.

[19]N. P. Jasti, A. S. Chouhan, and S. Avasthi, "Electron transport layer free perovskite solar cell with high open circuit voltage and dependence of scan rate on device performance", Israel Vaccum Society conference 2018, 2018.

[20]K. Kumari and S. Avasthi, "Laser crystallization of amorphous Ge thin films using pulsed Nd:YAG", 4th IEEE International Conference for Emerging Electronics 2018, 2018.

[21]R. Mukherjee, P. Singh, S. K. Podapangi, and S. Avasthi, "Improvement in morphology and carrier lifetime by inclusion of magnesium in the perovskite matrix", 4th International Conference on Perovskite Solar Cells and Optoelectronics, 2018.

[22]P. Ravindra, R. Ranjan, S. Raghavan, and S. Avasthi, "Control of conductivity in manganese vanadium oxide", 4th IEEE International Conference for Emerging Electronics 2018, 2018.

[23]V. R. S. Sandilya and S. Avasthi, "Pulsed laser deposition of vanadium oxide films", 2018 4th IEEE International Conference for Emerging Electronics (ICEE), 2018.

[24]A. Singh, A. S. Chouhan, and S. Avasthi, "Highly crystalline methylammonium tin iodide perovskite films using post-deposition methylamine vapor annealing for improved grain size", 2018 Spring Meeting of Material Research Society, 2018.

[25]P. Singh, R. Mukherjee, and S. Avasthi, "Enhancement in open-circuit voltage of semi-transparent, stable $MAPbI_{3-x}Br_x$ perovskite solar cells by methyl amine treatment and optimal hole transport layer", 4th IEEE International Conference for Emerging Electronics 2018, 2018.

[26]P. Singh, R. Mukherjee, and S. Avasthi, "Acetamidinium substituted methyl ammonium perovskite cells with higher open-circuit voltage and improved intrinsic stability", 2018 MRS Fall Meeting and Exhibit, 2018.

[27]S. Chaurasia, A. S. Chouhan, S. Raghavan, and S. Avasthi, "High-quality GaAs (100) thin films on silicon (100) using epitaxial germanium (100) buffer for low-cost III-V solar cells", 33rd European Photovoltaic Solar Energy Conference and Exhibition, 2017.

[28]S. Chaurasia, S. Raghavan, and S. Avasthi, "Characterization of titanium nitride as iron diffusion barrier for GaAs thin film solar cell on steel", 2017 Spring Meeting of Material Research Society, 2017.

[29]S. Chaurasia, S. Raghavan, and S. Avasthi, "Liquid phase epitaxy for growth of high-quality crystalline germanium (100) on silicon (100) wafers", 2017 Spring Meeting of Material Research Society, 2017.

[30]S. Chaurasia, S. Raghavan, and S. Avasthi, "High quality epitaxial germanium on Si(111) using a two- step annealing process", International Conference on Thin Films (ICTF) 2017, 2017.

[31]A. S. Chouhan, N. P. Jasti, and S. Avasthi, "Charge carrier lifetime in $CH_3NH_3PbI_3$ thin film: role of humidity", 33rd European Photovoltaic Solar Energy Conference and Exhibition, 2017.

[32]A. S. Chouhan, N. Prathibha J., R. B. Iyer, and S. Avasthi, "Compact-TiO_2 deposited via ALD for highly-repeatable and low-hysteresis perovskite solar cells", 2017 Spring Meeting of Material Research Society, 2017.

[33]K. Kumari and S. Avasthi, "Effects of columnar grain boundaries on thin film polycrystalline GaAs solar cells", 2017 Spring Meeting of Material Research Society, 2017.

[34]R. Mukherjee, P. Ravindra, and S. Avasthi, "Ultra-thin SiO_2 layers for defect-free si/oxide carrier selective contacts", 2017 Spring Meeting of Material Research Society, 2017.

[35]P. Ravindra and S. Avasthi, "Zinc molybdenum oxide - A new solar absorber", 2017 Spring Meeting of Material Research Society, 2017.

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[37]A. Singh, A. S. Chouhan, and S. Avasthi, "Interaction of humidity and polar molecules with mixed cation perovskite for high efficiency solar cells: materials and device study", International Conference on Thin Films (ICTF) 2017, 2017.

[38]A. Singh, S. Podapangi, A. S. Chouhan, and S. Avasthi, "Cation exchange for 3D-2H transformation of perovskite thin films using vapour annealing: an interplay of tolerance factor", International Symposium on Integrated Functionalities 2017, 2017.

[39]A. Chaudhary, N. Ravishankar, S. Avasthi, and S. Raghavan, "3D Ti/TiO_2 photoelectrodes for sensitized solar cells", 2016 IEEE 43rd Photovoltaic Specialist Conference (PVSC), 2016.

[40]S. Chaurasia, S. Raghavan, and S. Avasthi, "Ohmic contact formation between titanium nitride and n-type germanium thin films", International Conference of Young Researchers on Advanced Materials, International Union of Materials Research Society, Materials Research Society, 2016.

[41]S. Chaurasia, S. Raghavan, and S. Avasthi, "Epitaxial germanium films on silicon(100) for heterogeneous integration with silicon photonics", International Conference of Young Researchers on Advanced Materials, International Union of Materials Research Society, Materials Research Society, 2016.

[42]K. Kumari, A. Jayanthi, and S. Avasthi, "Effect of grain-boundaries on efficiency of thin-film gallium arsenide solar cells with columnar grain structure", International Conference of Young Researchers on Advanced Materials, International Union of Materials Research Society, Materials Research Society, 2016.

[43]P. Ravindra and S. Avasthi, "Electron-blocking properties of crystalline-silicon/Cu_2O heterojunctions for photovoltaics", 2016 Spring Meeting of Material Research Society, 2016.

[44]P. Ravindra and S. Avasthi, "Zinc-molybdenum-oxide: A new solar absorber", International Conference of Young Researchers on Advanced Materials, International Union of Materials Research Society, Materials Research Society, 2016.

[45]V. K. Singh and S. Avasthi, "Fabrication of esaki tunnel diode for silicon/pervoskite tandem cell", Conference on Emerging Materials 2016 (CEMAT-2016), 2016.

[46]V. K. Singh and S. Avasthi, "Tunnel diodes for perovskite/silicon tandem solar cell", International Conference of Young Researchers on Advanced Materials, International Union of Materials Research Society, Materials Research Society, 2016.

[47]S. Avasthi, W. E. McClain, G. Man, J. Jhaveri, K. A. Nagamatsu, A. Kahn, J. Schwartz, and J. C. Sturm, "Growth mechanism and carrier transport in hole-blocking TiO_2/silicon heterojunctions", 2014 Spring Meeting of Material Research Society, 2014.

[48]S. Avasthi, Y. McClain Afsar, G. Man, J. Jhaveri, K. A. Nagamatsu, A. Kahn, J. Schwartz, S. Wagner, and J. C. Sturm, "Hole-blocking metal-oxide/crystalline-silicon heterojunctions with recombination velocity of <100 cm/s", 2014 Spring Meeting of Material Research Society, 2014.

[49]J. Jhaveri, S. Avasthi, G. Man, K. A. Nagamatsu, W. E. McClain, J. Schwartz, A. Kahn, and J. C. Sturm, "Effect of annealing on stability of low interface recombination velocity at TiO_2/p-silicon interface", 2014 Spring Meeting of Material Research Society, 2014.

[50]K. A. Nagamatsu, J. Spechler, S. Avasthi, C. B. Arnold, and J. C. Sturm, "Silicon/organic heterojunction photovoltaic cell with 12.7% efficiency by use of spray-coated nanowire transparent conductor", 2014 Spring Meeting of Material Research Society, 2014.

[51]K. A. Nagamatsu, S. Avasthi, J. Jhaveri, and J. C. Sturm, "12% efficient silicon/PEDOT:PSS heterojunction solar cell fabricated at <100 °C", 39th IEEE Photovoltaic Specialists Conference, 2013.

[52]K. A. Nagamatsu, S. Avasthi, J. Spechler, C. B. Arnold, and J. C. Sturm, "Current mechanisms in silicon-organic heterojunction solar cells with transfer printed metallization", 2012 Fall Meeting of Material Research Society, 2012.

[53]S. Avasthi, S. Lee, Y.-L. Loo, and J. C. Sturm, "The role of P3HT barriers in P3HT-silicon heterojunction solar cells with a high open-circuit voltage", 2011 Spring Meeting of Material Research Society, 2011.

[54]S. Avasthi, Y. Qi, S. Ha, G. Vertelov, J. Schwartz, A. Kahn, and J. C. Sturm, "P3HT-silicon organic-silicon heterojunction for photovoltaic applications", 2010 Fall Meeting of Material Research Society, 2010.

[55]S. Avasthi, Y. Qi, G. Vertelov, J. Schwartz, A. Kahn, and J. C. Sturm, "Stability of electrical properties of silicon (100) surfaces passivated with 9,10-phenanthrenequinone", 2009 Fall Meeting of Material Research Society, 2009.

[56]J. C. Sturm, B. Hekmatshoar, L. Han, S. Avasthi, G. Vertelov, Y. Qi, J. Schwartz, A. Kahn, and S. Wagner, "Towards organic-based dielectrics for low-temperature silicon -based devices for large-area electronics", 2009 Fall Meeting of Material Research Society, 2009.

[57]S. Avasthi, Y. Qi, J. Schwartz, A. Kahn, and J. C. Sturm, "Electronic passivation of silicon (100) surfaces by organic layer of 9,10-phenanthrenequinone", 51st Electronic Materials Conference, 2007.

[58]S. Shankar, A. M. Tyryshkin, S. Avasthi, and S. A. Lyon, "Electron spin resonance of electrons in a large-area silicon MOSFET", APS March Meeting, 2007.

Invited Talks

[1]S. Avasthi, *Magic of scaling how we make nano scale devices?*, Indo–Taiwan Conference on Semiconductor Packaging Testing, Sastra University, Thanjavur, India, Jan. 2026.

[2]S. Avasthi, *Introduction to microfabrication for thz devices*, VEDA National Conference, Microwave Tube Research Development Centre (MTRDC), Bengaluru, India, Dec. 2025.

[3]S. Avasthi, *Why is semiconductor technology to important: magic of scaling?*, Reva University, Bengaluru, India, July 2025.

[4]S. Avasthi, *Magic of scaling how we make nano scale devices?*, international conference on Photophysics and Photochemistry (ICOPP-2024), IIT Bombay, Mumbai, India, Oct. 2024.

[5]S. Avasthi, *Magic of scaling how we make nano scale devices?*, international conference on Photophysics and Photochemistry (ICOPP-2024), IIT Bombay, Mumbai, India, Oct. 2024.

[6]S. Avasthi, A. A. Raj, S. Mandal, N. Anadkat, and K. Narasimhan, *Carrier lifetime, mobility and defects in perovskite crystals thin-films*, Technologist Industrialist Meet and Expo for Energy (TIMEE - 2024), CSIR-CSIO Chennai, India, June 2024.

[7]S. Avasthi, S. Patil, K. L. Narasimhan, P. Ramamurthy, A. Sadhanala, A. A. Raj, S. Mandal, and N. Anadkat, *Bulk and interface effects in perovskite solar cells*, Recent Advancements in Organic and Nanoelectronics (RAONE-2023), Oct. 2023.

[8]S. Avasthi and S. Kumar, *Perovskite and organic thin-film solar cells on steel*, SCDT-FlexE Webinar, June 2023.

[9]S. Avasthi, S. Patil, K. L. Narasimhan, P. Ramamurthy, A. Sadhanala, A. A. Raj, S. Mandal, and N. Anadkat, *Advanced electronics methods to characterize bulk and interface effects in perovskite solar cells*, National Conference on Modern Functional Materials (NCMFM-2023), Mar. 2023.

[10]S. Avasthi, *Are thin-film photovoltaics the future of solar energy?*, Institute Lecture, 2023.

[11]S. Avasthi, *Perovskite solar cells limited by interface: true or false?*, Advanced and New Generation Materials in Electrical and Electronics Engineering, Feb. 2021.

[12]S. Avasthi, *Solar cells: fundamentals, current trends, and future*, Recent Trends in Materials for Next Generation Applications, Feb. 2021.

[13]S. Avasthi, *Ensuring safety in academia*, Webinar on Environment, Health and Safety by the National Biopharma Mission, Nov. 2020.

[14]S. Avasthi, *Tuning composition for perovskite solar cells*, Institute Lecture, Reva University, Sept. 2020.

[15]S. Avasthi, *Perovskite solar cells*, National Conference on Micro Systems Technologies (NCMST 2020), Mar. 2020.

[16]S. Avasthi, *Tuning composition for perovskite solar cells*, Workshop on Energy, Materials and Devices, Center for Nano and Soft Matter Science, CeNS, Dec. 2019.

[17]S. Avasthi, *Ensuring safety in a multi-user facility*, Micro-Nano Fabrication and Characterization Symposium (MNFACS), July 2019.

[18]S. Avasthi, *Perovskite solar cells - towards commercialization*, 30th Annual General Meeting of MRSI and the First Indian Materials Conclave, Indian Institute of Science, Feb. 2019.

[19]S. Avasthi, *Perovskite solar cells - towards commercialization*, International Seminar On Recent Trends In Nano Science And Nanophotonics (NSNP 2019), K.E. College, Mahatma Gandhi University, Jan. 2019.

[20]S. Avasthi, *Perovskite solar cells - towards commercialization*, Bringing The Nanoworld Together 2018, Oxford Instruments, Nov. 2018.

[21]S. Avasthi, *Perovskite solar cells - towards commercialization*, Workshop on novel technologies to address needs of India, IISc, Oct. 2018.

[22]S. Avasthi, *Low-cost heterojunction solar cells*, IEEE Workshop on Nanotechnology and Sensors, IISc, Aug. 2018.

[23]S. Avasthi, *Perovskite solar cells*, IEEE EDS/SSCS Mini Colloquium on Solar Photovoltaics, IISc, Jan. 2018.

[24]S. Avasthi, *Ensuring safety: protection systems and user behaviour*, INAE-DST Consultive Meeting On Lab Safety, 2018.

[25]S. Avasthi, *Low-cost heterojunction solar cells*, 1st NTU-IISc joint symposium, IISc, Apr. 2017.

[26]S. Avasthi, *Towards roll-to-roll perovskite solar cells*, RD Activities on Solar Photovoltaics in India, IITB, Apr. 2017.

[27]S. Avasthi, *Solar cells (photovoltaics)*, Recent Advancement in Materials and Technology in Renewables, BMS College, Mar. 2017.

[28]S. Avasthi, *Morphology of perovskite thin-films*, Interaction Meeting of Indo-US R D Networked Joint Centre, JNCASR, Mar. 2017.

[29]S. Avasthi, *Challenges of growing highly-doped amorphous si layers using pecvd*, Bringing the Nanoworld Together 2016 organized, Oxford Instruments, Nov. 2016.

[30]S. Avasthi, *Heterojunction solar cells*, Chemical Engineering, Indian Institute of Science, Bengaluru, India, Bengaluru, India, Oct. 2016.

[31]S. Avasthi, *Heterojunction solar cells*, Invited talk at Unilever Inc. Aug. 2016.

[32]S. Avasthi, *Effect of solvent processing on the morphology of perovskite cells*, Workshop on Perovskite Semiconductor Solar Cells, IIT Bombay, May 2016.

[33]S. Avasthi, *Photovoltaics: silicon heterojunctions*, Workshop on Nanomaterials for Energy Applications, Anna University Regional Campus, Feb. 2016.

[34]S. Avasthi, *Microfabrication techniques*, Invited talk at Jain University, Jan. 2016.

Faculty Development Talks and Tutorials

[1]S. Avasthi, *Solar cells: past, present and future: thin-film photovoltaics tandem solar cells*, 7th International Conference on Emerging Electronics (ICEE 2025), Benglauru, India, Dec. 2025.

[2]S. Avasthi, *Magic of scaling how we make nano scale devices?*, FDP: Recent advancement on Organic and Nanoelectronics (RAONE-2025), SRM Institute of Science and Technology (SRMIST), Chennai, India, Oct. 2025.

[3]S. Avasthi, *Hole-conducting copper oxide using cvd for next-generation: thin film transistors, memristors, and gas sensors*, FDP: Advancements in Semiconductors, System Software, and AI for Digital Agriculture, JSS Academy of Technical Education, Benglauru, India, Sept. 2025.

[4]S. Avasthi, *Introduction to microfabrication for thz devices*, Tutorial on Sub-Terahertz Technologies, Microwave Tube Research Development Centre (MTRDC), Bengaluru, India, Sept. 2024.

[5]S. Avasthi, *P-type oxide devices and their application in beol tfts, microelectronics, sensors, and power devices*, NXP India Innovation Conclave, Bengaluru, India, Mar. 2024.

[6]S. Avasthi, *Fundamentals of micro-nanofabrication*, Bengaluru India Nano 2022, Mar. 2022.

[7]S. Avasthi, *Tuning composition for perovskite solar cells*, INUP-i2i 2nd Online Familiarization Workshop, IIT Guwahati, India, Feb. 2022.

[8]S. Avasthi, *Low-cost heterojunction solar cells: perovskite thin-film solar cells*, FDP on New Generation Solar Cell BNMIT, Bangalore, Oct. 2020.

[9]S. Avasthi, *Low-cost heterojunction solar cells: perovskite thin-film solar cells*, Special Workshop for PG students of physics and chemistry, KSTA, Jan. 2020.

[10]S. Avasthi, *Low-cost heterojunction solar cells*, MHRD Basic Training Program, CeNSE, Indian Institute of Science, Bengaluru, India, Oct. 2019.

[11]S. Avasthi, *Low-cost heterojunction solar cells*, MHRD Basic Training Program, CeNSE, Indian Institute of Science, Bengaluru, India, Mar. 2019.

[12]S. Avasthi, *Nano fabrication technologies*, Residential Training Program on Nanotechnology Relevant to Medical Research (RGUHS), CeNSE, Indian Institute of Science, Bengaluru, India, Mar. 2019.

[13]S. Avasthi, *Nano fabrication technologies*, IEEE Workshop on Nano-Enabled IoT, CeNSE, Indian Institute of Science, Bengaluru, India, Mar. 2019.

[14]S. Avasthi, *Microfabrication techniques part-i*, Tutorial at 10th Bengaluru INDIA NANO Conference, Bengaluru, India, Dec. 2018.

[15]S. Avasthi, *Low-cost heterojunction solar cells*, INUP Familiarization Training Workshop, Indian Institute of Science, Bengaluru, India, Oct. 2018.

[16]S. Avasthi, *Low-cost heterojunction solar cells*, Ministry of External Affairs International Workshop on Nanofabrication Technologies, Indian Institute of Science, Bengaluru, India, Aug. 2018.

[17]S. Avasthi, *Silicon photovoltaics*, SES Training Program, Indian Institute of Science, Bengaluru, India, Mar. 2018.

[18]S. Avasthi, *Perovskite solar cells*, INUP Familiarization Training Workshop, Indian Institute of Science, Bengaluru, India, Mar. 2018.

[19]S. Avasthi, *Photovoltaics: silicon heterojunctions*, INUP Outstation Workshop, IIT Roorkee, Roorkee, India, Apr. 2017.

[20]S. Avasthi, *Low-cost heterojunction solar cells*, FDP on Recent Advances in Engineering Physics, R V College, Bengaluru, India, Mar. 2017.

[21]S. Avasthi, *Photovoltaics: silicon heterojunctions*, INUP Familiarization Training Workshop, Indian Institute of Science, Bengaluru, India, Jan. 2017.

[22]S. Avasthi, *Photovoltaics: silicon heterojunctions*, INUP Familiarization Workshop, Indian Institute of Science, Bengaluru, India, Nov. 2016.

[23]S. Avasthi, *Heterojunction solar cells*, INUP Outstation Workshop, NIT Jaipur, Jaipur, India, Aug. 2016.

[24]S. Avasthi, *Solar cells*, FDP on Materials in Engineering & Technology, BMS College, Bengaluru, India, July 2016.

[25]S. Avasthi, *Effect of solvent processing on the morphology of perovskite cells*, National Symposium on Nano Science and Technology 2016, Indian Institute of Science, Bengaluru, India, June 2016.

[26]S. Avasthi, *Microfabrication techniques*, Nano for the Young Program, 8th Bangalore INDIA NANO Conference, Bengaluru, India, Mar. 2016.

[27]S. Avasthi, *Heterojunction solar cells*, INUP Outstation Workshop, NIT Agartala, Agartala, India, Feb. 2016.

[28]S. Avasthi, *Solar cell (photovoltaics)*, Special Training Course on Nanoelectronics for Scientists from India-Neighboring Countries, Indian Institute of Science, Bengaluru, India, Jan. 2016.

[29]S. Avasthi, *Semiconductor fabrication processes*, IEEE Student Chapter Workshop, Indian Institute of Science, Bengaluru, India, Oct. 2015.

[30]S. Avasthi, *Solar cells (photovoltaics)*, INUP Outstation Workshop, NIT Patna, Patna, India, Oct. 2015.

[31]S. Avasthi, *Solar cells (photovoltaics)*, INUP Familiarization Workshop, Indian Institute of Science, Bengaluru, India, July 2015.

[32]S. Avasthi, *Solar cells (photovoltaics)*, INUP Outstation Workshop, NIT Hamirpur, Hamirpur, India, Mar. 2015.

[33]S. Avasthi, *How to fabricate and test si solar cells*, 2nd IEEE International Conference on Emerging Electronics (ICEE)-2014, Bengaluru, India, Dec. 2014.

[34]S. Avasthi, *Solar cell*, INUP Workshop on Nanofabrication Technologies, Indian Institute of Science, Bengaluru, India, May 2014.

Book and Book Chapters

[1]U. Kumar and P. Sonkar, "Smart materials for engineering and sciences: current trends and future prospective scope of the book", in (Sept. 2022).

[2]D. Ginley et al., "Sustainable Photovoltaics", in *Solar Energy Research Institute for India and the United States (SERIIUS): Lessons and Results from a Binational Consortium* (Springer International Publishing, Feb. 2020), pp. 25–85, 10.1007/978-3-030-33184-9_2.

Patents & Know-how

	Title	Inventors	Date	Details
1	Deformable liquid semiconductor device and method of fabrication thereof	Anshu Pandey, Sushobhan Avasthi	14Aug2025	India Patent App. 202541077773
2	A method, apparatus, and system for a novel broadband self-powered Sb_2Te_3 and $MoTe_2$ heterostructure photodetector for infrared detection	Ashutosh Vishwakarma, Sushobhan Avasthi, Vijeyan C, Srinivasan Raghavan	28May2025	India Patent App. 202541051707
3	Non-contact temperature measurements of in-process Silicon wafer using broadband spectrally shaped source and a power detector.	Supradeepa V.R., Sushobhan Avasthi, Abhigyan Goswami, Sarthak Dash	05Mar2025	Invention Disclosure
4	Diketopyrrolopyrrole polymer and organic electronic device using the same	Anuj Kumar Palariya, Dipanjan Giri, Satish Patil, Sushobhan Avasthi	26Sep2024	India Patent 565364
5	Near IR Laser based contactless wafer temperature measurement	R. S. Patil, M. M. Kumar, V. K. Badam, et al.	14Mar2024	US Patent App. 63/565,249
6	Pyroelectric IR photodetector for Fire Sensing	Sushobhan Avasthi, Manoj Singh	27Sep2023	IP Docket IDR-CeN-2023-108
7	ICU Ventilator	Sushobhan Avasthi, Saurabh Chandorkar, Prosenjit Sen, Srinivasan Raghavan, Harshvardhan Gupta, Ankit Rao, Krishna Prasad, Hariharan Varadarajan, Jitthu Joseph, Swapnil More	27May2021	IP Docket IDR-CeN-2021-039
8	Solar cell	S. Chaurasia, S. Raghavan, & N. Bhat	06Mar2017	India Patent 305392
	–	–	18Oct2018	WIPO App. WO2018189752A1
9	Photovoltaic device and method of making the same	J. C. Sturm & J. Schwartz	15Dec2011	US App. US20110303904A1
	–	–	24Feb2016	Japan Patent JP5868963B2

	–	–	2012	Taiwan Patent 100118094
	–	–	12Jan2012	WIPO App. WO2011149850A3
	–	–	26Aug2013	Korea App. KR20130094728A
	–	–	10Apr2013	EU App. EP2577765A2
10	Hole-blocking silicon/titanium-oxide heterojunction for silicon photovoltaics	J. C. Sturm, W. E. McClain, J. Schwartz	19Sep2013	WIPO App. WO2013138635A1
	–	–	05Feb2015	US App. US20150034159A1
	–	–	04Nov2015	EU App. EP2826070A4
	–	–	05Feb2015	Japan App. JP2015514305A
	–	–	08Jan2015	Korea App. KR20150003181A
11	Silicon/organic heterojunction (SOH) solar cell and roll-to-roll fabrication process	Y. Huang, J. C. Sturm & K. Nagamatsu	15Nov2012	US Patent App. US20120285521A1

Academic Guidance

	In Progress	Graduated
Ph.D. Students	13	10
M.Tech. & M.Sc. (Engg.)	1	10
UG Final Projects	0	4
Post-Doctoral Fellows	3	8

Leadership

2023	**Member, AICTE Committee for Model Curricula for new UG programs**
	Part of the AICTE committee constituted to establish the model curricula for the new UG programs in VLSI and new diploma program in IC manufacturing. Also part of the AICTE delegation that visited Taiwan to explore joint collaborations.
2022	**Technical Chair, ICEE 2022**
	Headed the technical program committee that curated and reviewed the technical program. Compiled and edited the conference proceedings.
2019–Pres.	**Chair, IISc Institute Safety Committee**
	Oversee the Office of Laboratory Safety and Environmental Health (OLSEH), the regulatory body that regulates safety in all laboratories on the IISc campus.
2018	**Technical Program Committee, Lead Photovoltaic Track, ICEE 2018**
	Curated the technical program of the PV track in the conference. Reviewed the technical papers.

Grants

	Title	Role	Program, Agency	Amount (₹ 00,000)	Duration
1	Design and Technology Development of Solid State X- ray Detector for Astronomical Imaging	PI	ISRO	658.9	2026-28
2	Development of Pyroelectric IR sensor for fire detection	PI	DIAR-COE, DRDO	162.8	2025-27
3	Investigations of Ge-Sn alloys suitable for photodetector in SWIR range	co-PI	ERIP/ER, DRDO	38.3	2024-27
4	A Project Proposal for Research and Application in Micro and Nanotechnologies (PRAMAN)	Chief-I	MeitY	9782.0	2024-28
5	LAM Research India- IISc Training Program on SEMulator3D	PI	MeitY	1379.6	2024-26
6	Semiconductor Fabrication & Characterization Training for Students from Tribal Community	PI	MoTA	1323.0	2024-27
7	MoOClx gas concentration sensors	co-PI	Lam Research	58.1	2024-26
8	Ultrasonic Gas Mixing	co-PI	Lam Research	42.5	2023
9	Flexible and high-efficiency organic-perovskite tandem solar cells	PI	Solar Challenge, DST	479.3	2023-25
10	Singlet-fission enhanced silicon solar cells	PI	SERB, DST	50.0	2023-25
11	Flexible photovoltaic minimodules based on lead halide perovskite technology	co-PI	DIAR-COE, DRDO	209.2	2023-26
12	Spatial And Time Resolved Temperature Measurement On Silicon Wafer	PI	Lam Research	19.0	2022-23
13	Mid-IR photodetectors based on colloidal I-III-VI2 quantum dots	PI	Antigone Solutions	63.5	2022-23
14	Low-cost ICU Ventilator	PI	Western Digital	40.0	2021-22
15	Indian Nanoelectronics Users' Programme - Idea to Innovation (INUP-i2i)	co-PI	MeitY	1710.0	2021-22
16	Nanostructured materials for hyperspectral sensing	co-PI	IRPHA, DST	277.3	2020-24
17	Optimization of crystal oscillator fabrication method to improve yield	PI	Rakon India	10.0	2020-22
18	Pb-free perovskite for commercially deployable high-efficiency thin-film photovoltaics	PI	STARS, MHRD	84.0	2020-22
19	Large-area and stable perovskite solar cells	PI	SERB, DST	77.1	2020-22
20	New Barrier Encapsulation for Flexible Energy Storage and PV Devices	co-PI	AISRF, DST	35.0	2019-21
21	Nanoelectronics Network For Research and Application (NNetRA)	co-CI	DST, MoE, & MeitY	7500.0	2018-23
22	Facility argumentation at CeNSE	co-PI	FIST, DST	252	2018-23
23	Ge-on-Steel for Flexible Thin-Film Solar Cells	PI	Synergy Projects, Office of PSA	230.80	2017-19
24	Hole-conducting copper (+1)-oxide semiconductors for thin-film transistors	PI	NSAG-I, DST	118.26	2017-20
25	Improvement in Solar efficiency	PI	BEL	15.0	2017-18
26	Crystalline-silicon/perovskite tandem solar cell for high-efficiency photovoltaics	PI	SERB, DST	54.98	2014-18
27	Solar Energy Research Institute for India and the United States (SERIIUS), Task PV3	Co-PI	IUSSTF, DST	$\sim$ 625	2013-17
28	Crystalline-silicon/perovskite tandem solar cell for high-efficiency photovoltaics	PI	Start-up, IISc	36.0	2014-15